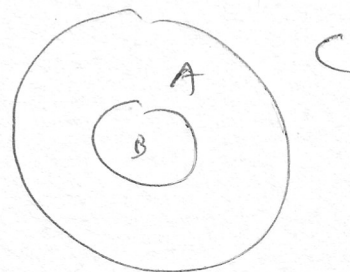


Configurations with hole inside.

For presentation later, label regions as

$$\hat{\rho}_A = \text{Tr}_B (\hat{\rho}_{AB}) \quad \dots (8.1)$$



$$\rho_{AB} = \text{Tr}_C (\hat{\rho}) \quad \dots (8.2)$$

We start from previous result for $\hat{\rho}_{AB}$, then obtain ρ_A

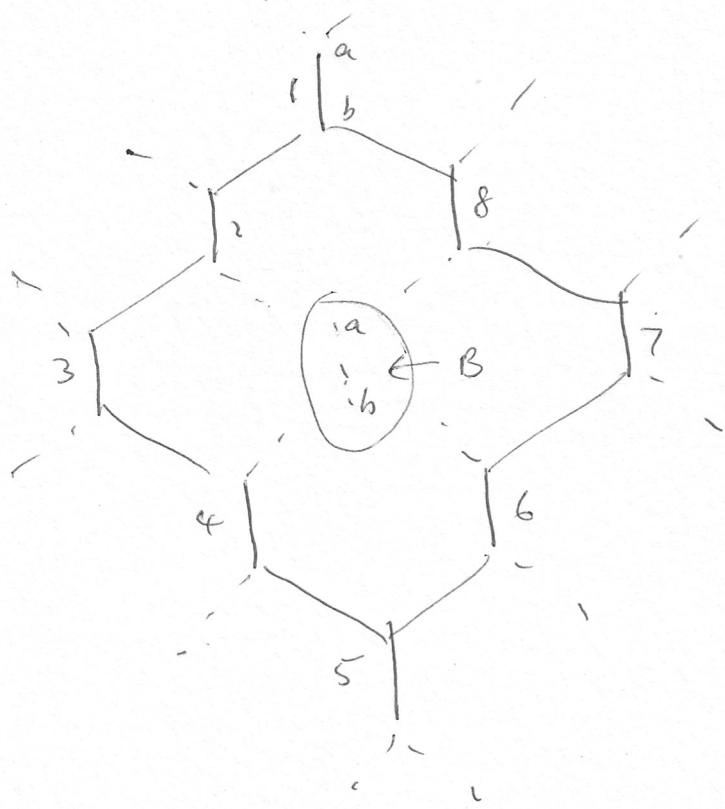
From known results of topological entanglement entropy:

exp

$$S_{AB} = (L_{AB} - 1) \ln 2 + S_{F, AB}^0 \quad \dots (8.3)$$

$$S_A = ((L_{AB} - 1) + (L_A - 1)) \ln 2 + S_{F, A}^0 \quad \dots (8.4)$$

Example:



A has 16 sites
B has 2 sites

$L_{int} = 6 \quad L_{in} = 2$

cut-bonds

$$\begin{aligned} 2L_{int} &= 12 \\ 2L_{in} &= 4 \end{aligned}$$

Naive counting:

A has Hilbert dimension 2^8 , spin dimension 2^{16}
 number of loops = 1

$$\hat{P}_{FA}^0 = \frac{1}{2^8} (1 + (icc)icc + \dots) \quad (8.1)$$

$$\hat{P}_{S,A} = \frac{1}{2^7} \left(\frac{1+W}{2} \right) \left\{ \frac{1}{2^8} (1 + \dots) \right\} \quad (8.2)$$

with $\text{Tr } \hat{P}_{FA}^0 = 1 \quad \text{Tr } \hat{P}_{S,A} = 1$

so
$$\hat{P}_{S,A} = \frac{1}{2^7} \left(\frac{1+W}{2} \right) \hat{P}_{FA}^0 \quad (8.3)$$

and
$$S_A = 7 \ln 2 + S_{F,A}^0 \quad (8.4)$$

(8.4) is wrong, since $(n-1) + (m-1) = 6 \neq 7$ (18.3)

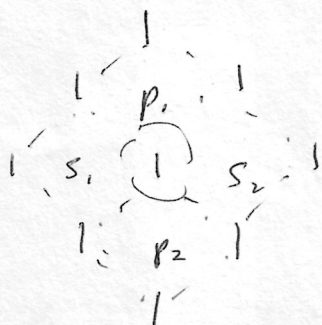
So (8.3) and (8.2) must be wrong.

Why?

Issue easy to resolve in dimer limit.

where each 2-dimer is an effective spin,

A has 8 composite spin



$$\hat{O} = \begin{pmatrix} \Sigma_x & \Sigma_y \\ \Sigma_y & \Sigma_x \end{pmatrix}$$

We note $O_{p1} O_{p2}$ is an operator entirely on A

$O_{s1} O_{s2}$ is on B

$$O_{p1} O_{p2} O_{s1} O_{s2} = W$$

$$\hat{\rho}_A = \frac{1}{2^6} \left(\frac{1+W}{2} \right) \left(1 + \frac{O_{p1} O_{p2}}{2} \right) \quad (8.5)$$

correctly gives $S_A = 6 \ln 2$

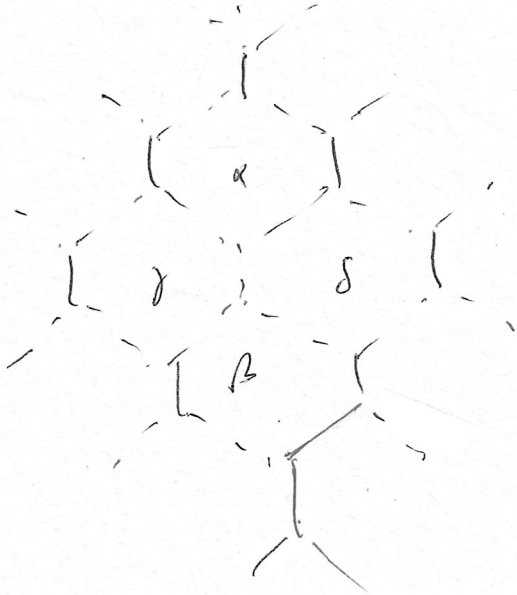
In original language for Kitaev model,

(AB) has 18

spins and four loops,

with dimer $(\sigma^z \sigma^z)$

within each unit cell.



$$\hat{\rho}_{AB} = \frac{1}{2^6} \frac{(1+W_\alpha)(1+W_\beta)(1+W_\gamma)(1+W_\delta)}{2^4} \times \left\{ \frac{1}{2^8} \left(1 + \underbrace{\langle \sigma_{1a}^z \sigma_{1b}^z \rangle}_1 \sigma_{1a}^z \sigma_{1b}^z + \dots \right) \right\}$$

note $W_\alpha W_\beta W_\gamma W_\delta = W$ identity in A and

$$W_\alpha W_\beta = W_\gamma W_\delta W$$

$$= \frac{1}{2^6} \left(\frac{1+W}{2} \right) \left[\frac{1 + W_\alpha + W_\beta + \dots + W_\alpha W_\beta + W_\alpha W_\gamma + W_\alpha W_\delta}{2^3} \right] \times \left[\frac{1}{2^8} (1 + \sigma_{1a}^z \sigma_{1b}^z + (\sigma_{2a}^z \sigma_{2b}^z) + \dots + \dots) \right] \quad \text{B.6)$$

note: for pairs, $W_\beta W_\gamma = W_\alpha W_\delta W$

so need not be included explicitly. similarly for $W_\beta W_\delta, W_\gamma W_\delta$

products of three, $W_\alpha W_\beta W_\gamma = W_\delta W$ also need not be included.

Perform trace over B to get ρ_A .

$$\frac{1}{2^4} \left\{ \frac{1}{2^2} [1 + w_a + w_b w_p] \left[1 + (\sigma_{Ba}^z \sigma_{Bb}^z) + \dots \right] \right\}$$

$\downarrow$ $\downarrow$
 1 $w_a' w_p'$

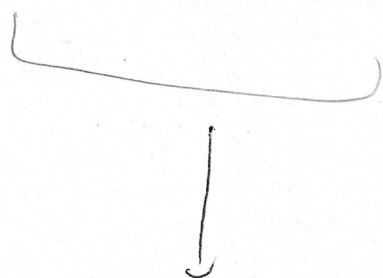
where

$$\begin{aligned} w_a' &\equiv w_a \sigma_{Ba}^z \\ w_p' &\equiv w_p \sigma_{Bb}^z \end{aligned}$$

entirely in A

$$\rho_A = \frac{1}{2^4} \left(\frac{1+w}{2} \right) \left(\frac{1+w_a' w_p'}{2} \right) \left\{ \frac{1}{2^2} \left[1 + \frac{(\sigma^z \sigma^z)}{2} + \dots \right] \right\}$$

dim in A



analogous to (8.5)

is correct EE.

$$\rho_A^0 \quad \dots (8.7)$$

but purely a projector

$$\left(\frac{1 + \sigma_{Ba}^z \sigma_{Bb}^z}{2} \right) \dots \left(\frac{1 + \sigma_{Ba}^z \sigma_{Bb}^z}{2} \right)$$

Trouble is that, if $\langle \vec{\sigma}_{Pa} \vec{\sigma}_{Pb} \rangle \neq 1$,

we get $\langle W_\alpha W_\beta \vec{\sigma}_{Pa} \vec{\sigma}_{Pb} \rangle W'_\alpha W'_\beta$

||

$$\langle \vec{\sigma}_{Pa} \vec{\sigma}_{Pb} \rangle \neq 1$$

and $(1 + \langle \vec{\sigma}_{Pa} \vec{\sigma}_{Pb} \rangle W'_\alpha W'_\beta)$ fails to be a projector.

and we won't have the factorization in (8.7)

We shall see that the resolution is, we get

$$\begin{aligned} & 1 \times (\text{dim operators}) \\ & + W'_\alpha W'_\beta \langle \vec{\sigma}_{Pa} \vec{\sigma}_{Pb} \rangle \times (\text{dim operators}) \end{aligned} \quad \Bigg) \quad (8.8)$$

These two lines can be rearrange to get

$$\begin{aligned} & (\text{dim operators})_e \leftarrow \text{for } W'_\alpha W'_\beta = 1 \\ & + (\text{dim operators})_o \leftarrow \text{for } W'_\alpha W'_\beta = -1 \end{aligned} \quad \Bigg) \quad (8.9)$$

with $()_e, ()_o$ orthogonal

and each $()_e, ()_o$

isomorphic to Majorana's

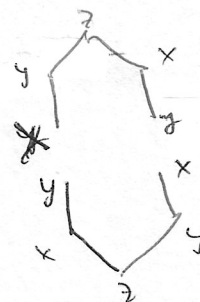
Hint for (8.8) : (8.8)

dimer operators $\longleftrightarrow$ Majorana's in A
in A

We note however that

$$W'_\alpha W'_\beta = \dots$$

is not a dimer operator in A



(The operators at the end points are "wrong":
They need to be $\frac{1}{2}$'s in order to be dimer operators)

However, we note that $(W'_\alpha W'_\beta)$ commutes with

all dimer operators in A (because they come from W_α, W_β)

$W'_\alpha W'_\beta$ can be diagonalized, with eigenvalues ± 1 . $\rightarrow$

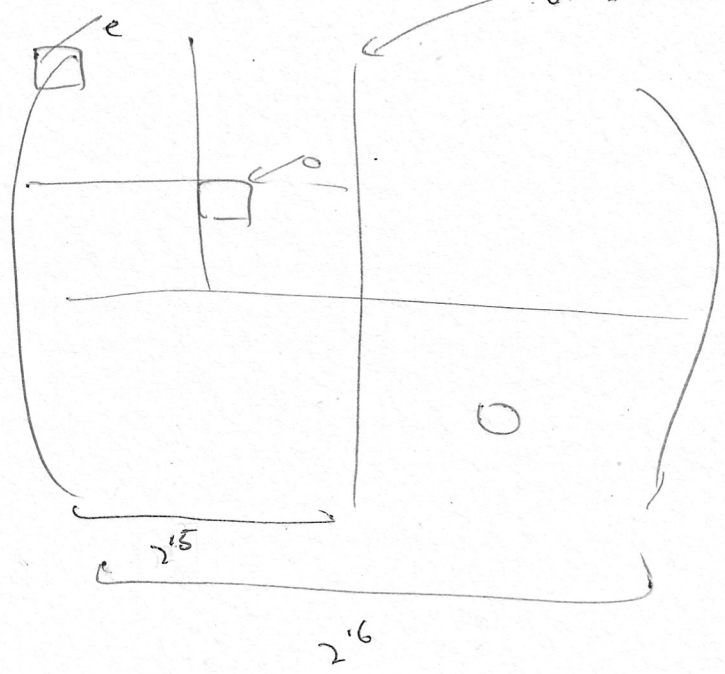
recombines to ± 1 sectors in (8.8)

even though the two sets of dimer operators
in (8.8) are not the same.

if

$\hat{P}A$ has the following structure $w=1$ sector

2^6 blocks repeated
 2^6 blocks repeated



(8.10)

$\square^e, \square^o$ each $2^8 \leftrightarrow$ 16 Majorana's

To see clearer tree structure, go back to general $\hat{P}_{AD}$
and proceed as in p. 8.4.

$$\hat{P}_A = \frac{1}{26} \left(\frac{1+W}{2} \right)^{\left(\frac{1}{24} \right)} \left[\frac{1}{4} \text{Tr} \left(\left(1 + W_\alpha + \dots + W_\alpha W_\beta + W_\alpha W_\gamma + W_\alpha W_\delta \right) \right) \right. \\ \left. - \left(1 + \langle X(\vec{\sigma}_{Ba} \vec{\sigma}_{BL}) \rangle + \langle X(\sigma_{Ba}^y \sigma_{BL}^y) \rangle + \langle \dots \rangle (\sigma_{Ba}^y \sigma_{BL}^x \sigma_{BL}^y \sigma_{BL}^y) \right) \right]$$

and look for finite trace.

note:

terms that do not involve any σ_A 's in $\{ \}$ give

finite contribution from

$$1 - 1 \text{ and } W_\alpha W_\beta \vec{\sigma}_{Ba} \vec{\sigma}_{BL} - i W'_\alpha W'_\beta \text{ coeff } \langle \vec{\sigma}_{Ba} \vec{\sigma}_{BL} \rangle$$

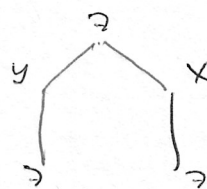
W_α can have a finite trace w.r.t

$$W'_\alpha \vec{\sigma}_{Ba}$$

$$\sigma_{Ba}^y \vec{\sigma}_{Ba} \sigma_{BL}^x$$

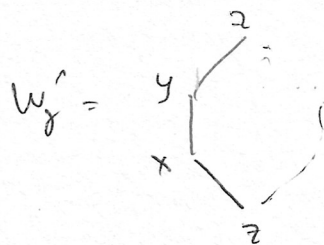
and this will produce

$$= \text{dim entirely} = A$$

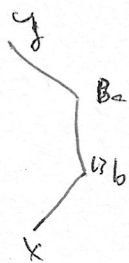


$\sigma_{Ba}^y, \sigma_{BL}^x$
"correct"
the end points
of W'_α

$$W_Y = W'_Y \sigma_{Ba}^x \sigma_{Bb}^y$$



to get first trace, can $\times$



and get back again



= dim $\sim A$

we saw that $W_\alpha W'_\beta \sim 1$ can $\sim (\sigma_{Ba}^z \sigma_{Bb}^z) \sim 2$

to get $W'_\alpha W'_\beta \in A$.

similarly $W_\alpha W_\beta \sim 1$ can

dim $\times \left[(\sigma_{Ba}^z \sigma_{Bb}^z) \times \text{any dimer in } A \right]$

to get $[W'_\alpha W'_\beta \neq \text{dimer in } A]$

$$\text{eg } \frac{1}{6} \text{Tr}_B \left(W_\alpha W_\beta \right) \quad \frac{\langle \sigma_{Ba}^z \sigma_{Bb}^z \sigma_{La}^z \sigma_{Lb}^z \rangle (\sigma_{Ra}^z \sigma_{Rb}^z \sigma_{La}^z \sigma_{Lb}^z)}{\text{Tr} \in 2}$$

$$\left(W'_\alpha W'_\beta \sigma_{La}^z \sigma_{Lb}^z \right) \langle \sigma_{Ra}^z \sigma_{Rb}^z \sigma_{Ba}^z \sigma_{Bb}^z \rangle$$

operators in A

but not dimer in A

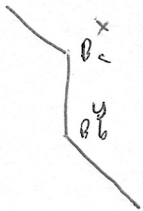
terms with only one σ_B in 2

cannot have finite trace with any factors in 1.

e.g. $(\sigma_{2b}^y \sigma_{3c}^y)$

none of the factors in 1 can generate σ_{2c}^y
without generating also σ_{3b} .

($W_x W_y$, W_s both contain σ_{3b})



can produce finite trace by

$\hookrightarrow W_x W_s$

but again reduces to zero in A.

(more systematic proof !!)

$$\tilde{\rho}_A = \frac{1}{2^b} \left(\frac{1+W}{2} \right) \left\{ \frac{1}{2^d} \left(1 + (W'_\alpha W'_\beta) \langle \vec{\sigma}_\alpha \vec{\sigma}_\beta \rangle \right. \right. \\ \left. \left. + \langle \vec{\sigma}_{1a} \vec{\sigma}_{1b} \rangle \sigma_{1a}^2 \sigma_{1b}^2 + \langle \sigma_{1a}^2 \sigma_{1b}^2 \sigma_{2a}^2 \sigma_{2b}^2 \rangle W'_\alpha W'_\beta \sigma_{1a}^2 \sigma_{1b}^2 \right. \right. \\ \left. \left. + \left(\begin{array}{c} \text{other terms} \\ \text{entirely in } A \end{array} \right) \right\}$$

$$+ \langle \vec{\sigma}_{B_1} \vec{\sigma}_{B_2} \times \text{dimens in } A \rangle (W'_\alpha W'_\beta \times \text{dimens in } A) \quad (8.11)$$

which is thus of the form claimed in (8.8).

Let $P = \vec{\sigma}_B \cdot \vec{\sigma}_A$, we can write

$$\tilde{\rho}_A = \frac{1}{2^b} \left(\frac{1+W}{2} \right) \left\{ \left(\frac{1+W'_\alpha W'_\beta}{2} \right) \left[\frac{1}{2^d} \left(\langle \frac{1+P}{2} \rangle + \langle \frac{1-P}{2} \rangle (\text{dimens}_1) \right) \text{dimens}_1 \right. \right. \\ \left. \left. + \langle \frac{1+P}{2} \rangle (\text{dimens}_1) (\text{dimens}_2) \times (\text{dimens}_1) (\text{dimens}_2) \right. \right. \\ \left. \left. + \dots \right] \right. \\ \left. + \left(\frac{1-W'_\alpha W'_\beta}{2} \right) \left[\frac{1}{2^d} \left(\langle \frac{1+P}{2} \rangle + \langle \frac{1-P}{2} \rangle (\text{dimens}_1) \right) \text{dimens}_1 \right. \right. \\ \left. \left. + \langle \frac{1-P}{2} \rangle (\text{dimens}_1) (\text{dimens}_2) \times (\text{dimens}_1) (\text{dimens}_2) \right. \right. \\ \left. \left. + \dots \right] \right\}$$

with $\text{dimens} \Leftarrow \text{dimens operator in } A$.